

DBMS LAB: SQL Joins and Transactions

Your environment has been set up for using Node.js 24.16.0 (x64) and npm.

```
C:\Users\sriha>cd C:\Program Files\MySQL\MySQL Server 8.0\bin
```

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
```

Enter password: ****

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 15

Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> CREATE DATABASE assignment_1;
Query OK, 1 row affected (0.03 sec)
```

```
mysql> USE assignment_1;
```

Database changed

```
mysql> CREATE TABLE roles (
  ->     role_id INT PRIMARY KEY,
  ->     role_name VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.10 sec)
```

```
mysql>
```

```
mysql> CREATE TABLE users (
  ->     user_id INT PRIMARY KEY,
  ->     name VARCHAR(50),
  ->     email VARCHAR(50),
  ->     role_id INT
  -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql>
```

```
mysql> CREATE TABLE departments (
  ->     dept_id INT PRIMARY KEY,
```

```
mysql>
mysql> CREATE TABLE departments (
  ->     dept_id INT PRIMARY KEY,
  ->     dept_name VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE employees (
  ->     emp_id INT PRIMARY KEY,
  ->     emp_name VARCHAR(50),
  ->     dept_id INT,
  ->     manager_id INT,
  ->     salary DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE customers (
  ->     customer_id INT PRIMARY KEY,
  ->     customer_name VARCHAR(50),
  ->     city VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE orders (
  ->     order_id INT PRIMARY KEY,
  ->     customer_id INT,
  ->     order_date DATE,
  ->     total_amount DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE accounts (
  ->     account_no INT PRIMARY KEY,
  ->     holder_name VARCHAR(50),
  ->     balance DECIMAL(10,2)
  -> );
```

```
->     holder_name VARCHAR(50),
->     balance DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.10 sec)

mysql>
mysql> SELECT emp_name, dept_name
-> FROM employees e
-> INNER JOIN departments d
-> ON e.dept_id = d.dept_id;
Empty set (0.03 sec)

mysql>
mysql> SELECT name, email, role_name
-> FROM users u
-> LEFT JOIN roles r
-> ON u.role_id = r.role_id;
Empty set (0.01 sec)

mysql>
mysql> SELECT name, role_name
-> FROM users u
-> RIGHT JOIN roles r
-> ON u.role_id = r.role_id;
Empty set (0.00 sec)

mysql>
mysql> SELECT name, email, role_name
-> FROM users u
-> LEFT JOIN roles r
-> ON u.role_id = r.role_id
-> UNION
-> SELECT name, email, role_name
-> FROM users u
-> RIGHT JOIN roles r
-> ON u.role_id = r.role_id;
Empty set (0.01 sec)

mysql>
mysql> SELECT e.emp_name AS Employee, m.emp_name AS Manager
-> FROM employees e
```

```
mysql> SELECT e.emp_name AS Employee, m.emp_name AS Manager
-> FROM employees e
-> LEFT JOIN employees m
-> ON e.manager_id = m.emp_id;
Empty set (0.01 sec)

mysql>
mysql> SELECT o.order_id, customer_name, city, total_amount
-> FROM customers c
-> JOIN orders o
-> ON c.customer_id = o.customer_id;
Empty set (0.01 sec)

mysql>
mysql> SELECT dept_name, COUNT(*), AVG(salary), MAX(salary)
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id = d.dept_id
-> GROUP BY dept_name;
Empty set (0.01 sec)

mysql>
mysql> SELECT emp_name, salary, dept_name
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id = d.dept_id
-> WHERE salary > 50000
-> AND dept_name = 'CSE';
Empty set (0.00 sec)

mysql>
mysql> SELECT customer_name, SUM(total_amount) AS TotalPurchase
-> FROM customers c
-> JOIN orders o
-> ON c.customer_id = o.customer_id
-> GROUP BY customer_name
-> ORDER BY TotalPurchase DESC;
Empty set (0.00 sec)

mysql>
mysql> SELECT student_name, course_name, faculty_name
```

```
mysql>
mysql> SELECT student_name, course_name, faculty_name
    -> FROM students s
    -> JOIN enrollments e
    -> ON s.student_id = e.student_id
    -> JOIN courses c
    -> ON e.course_id = c.course_id
    -> JOIN faculty f
    -> ON c.faculty_id = f.faculty_id;
ERROR 1146 (42S02): Table 'assignment_1.students' doesn't exist
mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO accounts VALUES (104, 'Anil', 25000);
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO accounts VALUES (105, 'Ravi', 10000);
Query OK, 1 row affected (0.00 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE accounts
    -> SET balance = balance - 5000
    -> WHERE account_no = 101;
Query OK, 0 rows affected (0.01 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```

```
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE accounts
      -> SET balance = balance - 3000
      -> WHERE account_no = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql>
mysql> UPDATE accounts
      -> SET balance = balance + 3000
      -> WHERE account_no = 999;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE accounts
      -> SET balance = balance + 2000
      -> WHERE account_no = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql>
mysql> SAVEPOINT s1;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> UPDATE accounts
    -> SET balance = balance - 1000
    -> WHERE account_no = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```

```
mysql>
mysql> ROLLBACK TO s1;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SHOW FULL TABLES;
+-----+-----+
| Tables_in_assignment_1 | Table_type |
+-----+-----+
| accounts                | BASE TABLE |
| customers               | BASE TABLE |
| departments             | BASE TABLE |
| employees               | BASE TABLE |
| orders                  | BASE TABLE |
| roles                   | BASE TABLE |
| users                   | BASE TABLE |
+-----+-----+
7 rows in set (0.04 sec)
```

```
mysql> |
```